

Employee Performance & Productivity

Full Workforce Analytics Report

100,000 employees · 9 departments · 7 job titles · hiring window 2014–2024

Avg monthly salary	\$6,403	Attrition rate	10.0%
Avg performance	3.00 / 5	High performers (4–5)	39.9%
Avg satisfaction	3.00 / 5	Avg tenure	4.5 years
Avg work hours	45.0 / week	Avg training	50 h

Prepared by the automated analytics pipeline · September 2026

Executive Summary

This report analyses the full employee performance and productivity dataset — 100,000 employee records across 9 departments and 7 job titles, covering hiring dates from September 2014 to September 2024. The data is complete (no missing values, no duplicate IDs) and supports descriptive analysis, correlation studies, equity testing and predictive modelling.

Headline findings

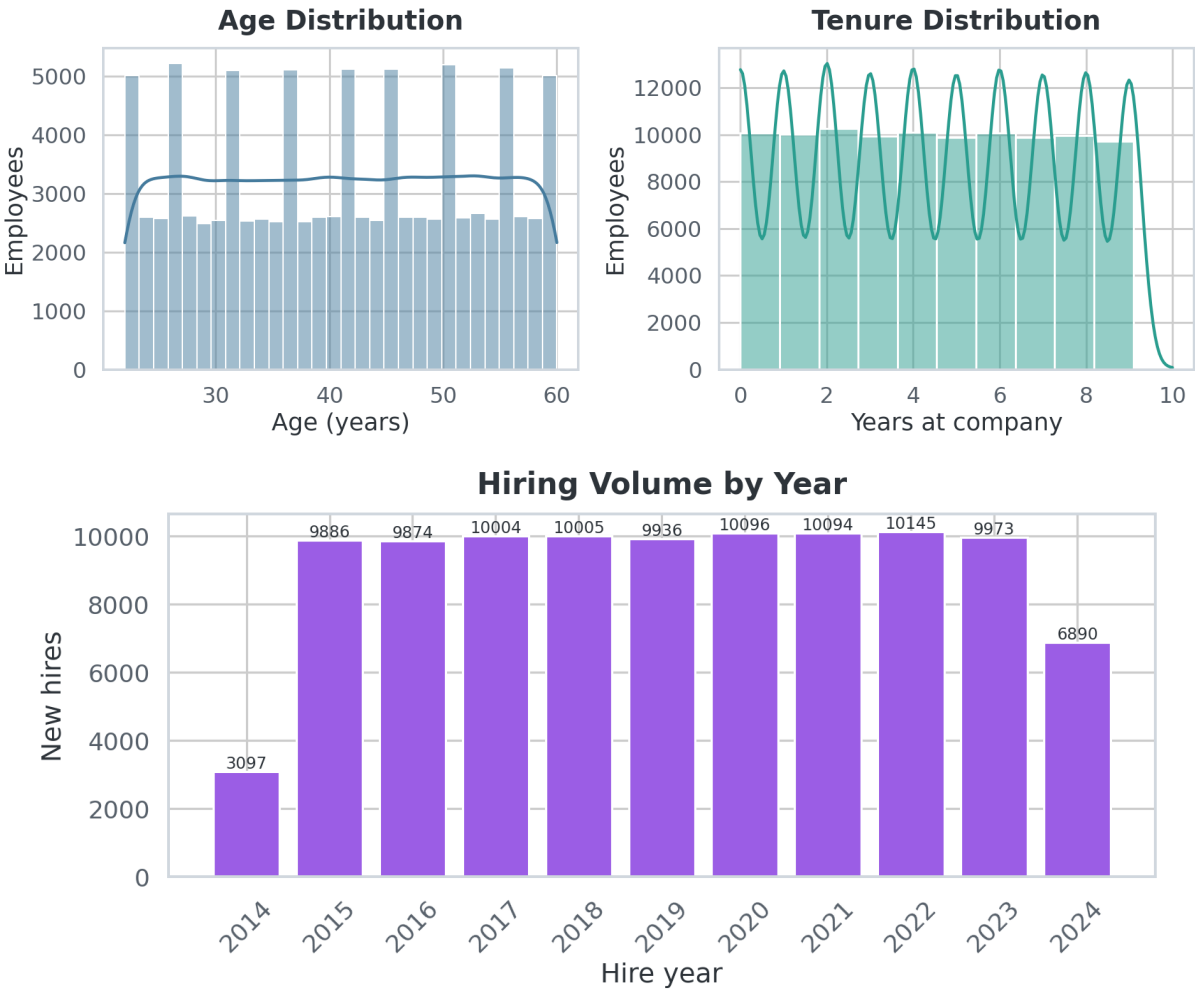
- **Pay equity is healthy.** Women earn on average \$4.50 *more* than men company-wide, and the within-title gap never exceeds $\pm\$25$ ($< 0.4\%$ of pay). No corrective compensation action is indicated.
- **Salary is almost perfectly explained by two factors.** A linear model on job title and performance score reaches $R^2 = 0.991$ ($MAE \approx \$95$). Engineers and Managers carry a $\approx +\$2,600$ /month title premium and every performance point adds $\approx +\$493$ /month. Department, education and gender contribute almost nothing.
- **Attrition is 10.0% and statistically unpredictable from this data.** Logistic regression and random forest both reach $AUC \approx 0.49$ — coin-flip discrimination — and every demographic, engagement and workload segment sits within 9.5–10.3% of the company rate. Departures in this dataset carry no observable signal.
- **Engagement levers show no measurable effect.** Training hours, overtime, sick days, team size and remote-work frequency correlate with performance and satisfaction at $|r| < 0.01$. Only salary tracks performance ($r = 0.51$), which is a definitional link (pay formula), not a causal one.
- **Departments are near-identical.** Headcount, salary, performance, satisfaction and attrition vary by less than 1.5% across all nine departments — an unusual uniformity.
- **Data quality caveat.** The independence of features, the near-perfect salary formula and the absence of any attrition signal strongly suggest a synthetically generated dataset. Findings should be read as a validated analysis methodology rather than organizational truth.

Recommendations focus on (1) preserving the current pay-equity position, (2) auditing the performance-measurement process given its central role in pay, and (3) improving attrition data capture — exit interviews and engagement surveys — because the current fields cannot identify flight-risk employees. Section 8 details the full recommendation set.

1. Data Overview & Quality

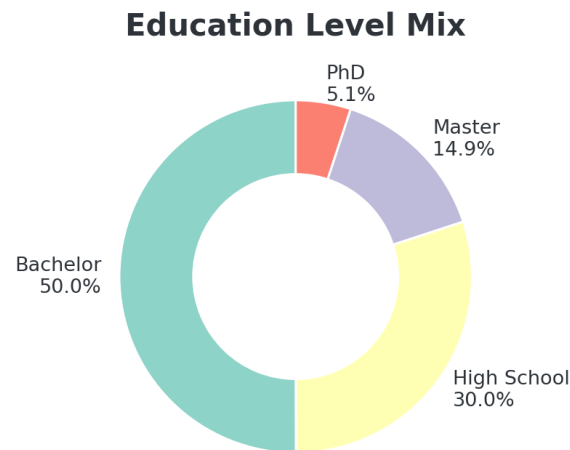
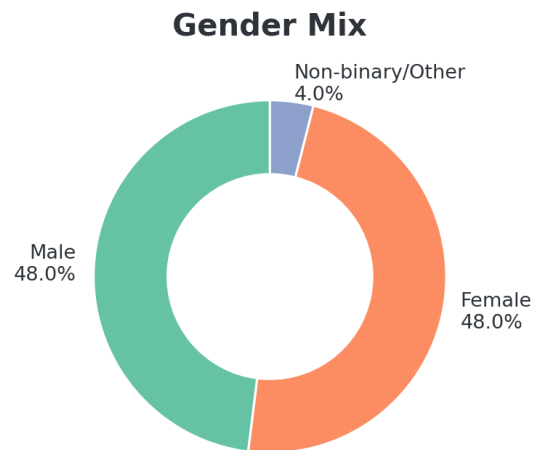
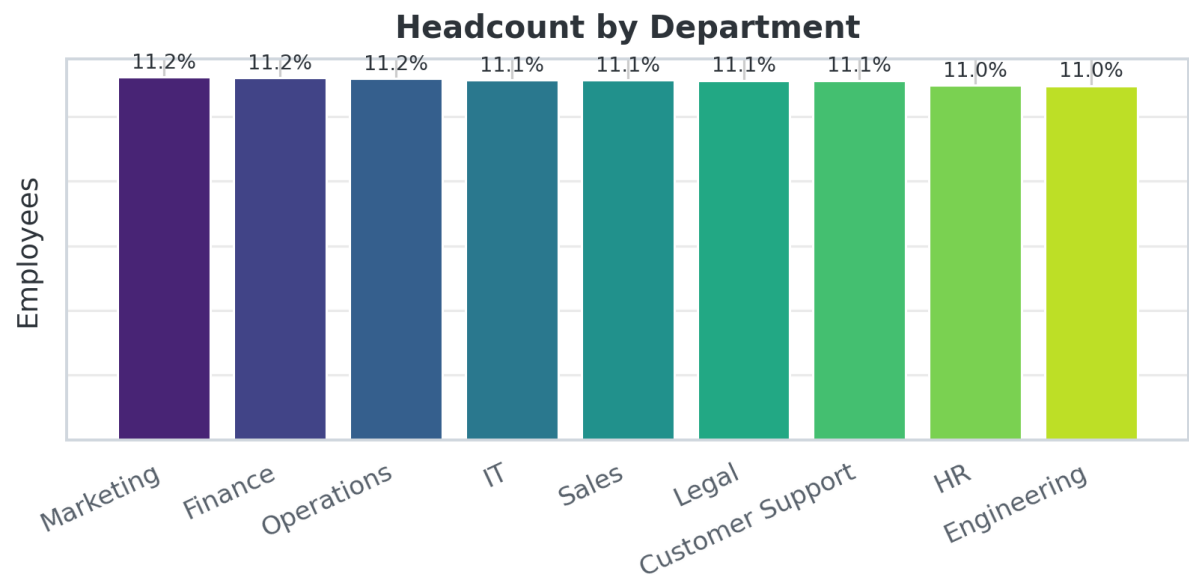
Property	Value
Records	100,000 employees
Fields	21 (13 numeric, 5 categorical, 1 boolean, 1 date)
Missing values	0
Duplicate Employee IDs	0
Departments	9 (each ≈ 11,000 staff)
Job titles	7 — Analyst, Consultant, Developer, Engineer, Manager, Specialist, Technician
Hire-date range	2014-09 → 2024-09

No cleaning was required: zero nulls, zero duplicate IDs, all ranges plausible (age 22–60, tenure 0–10 years, salary \$3,850–\$9,000, work hours 30–60). 'Other' gender values are reported as Non-binary/Other throughout.



2. Workforce Composition

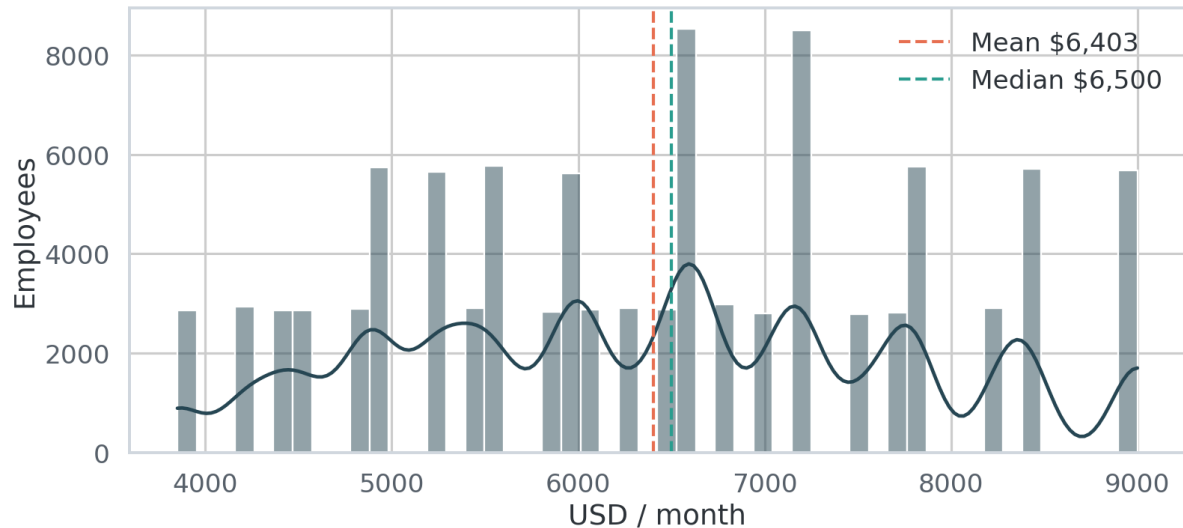
The workforce is evenly spread: every department employs between 10,956 (Engineering) and 11,216 (Marketing) staff. The gender mix is roughly balanced and education is dominated by Bachelor-level staff, with meaningful Master/PhD representation. Hiring volume is steady at 9,000–11,000 hires per year across the decade covered.



3. Compensation Analysis

Monthly salary averages \$6,403 (median \$6,500, $\sigma = \$1,373$) and is roughly normally distributed. Contrary to typical HR data, salary does not vary meaningfully by department (\$6,378–\$6,417) or education level. What does move pay is job title and performance score — confirmed by the regression in section 3.3.

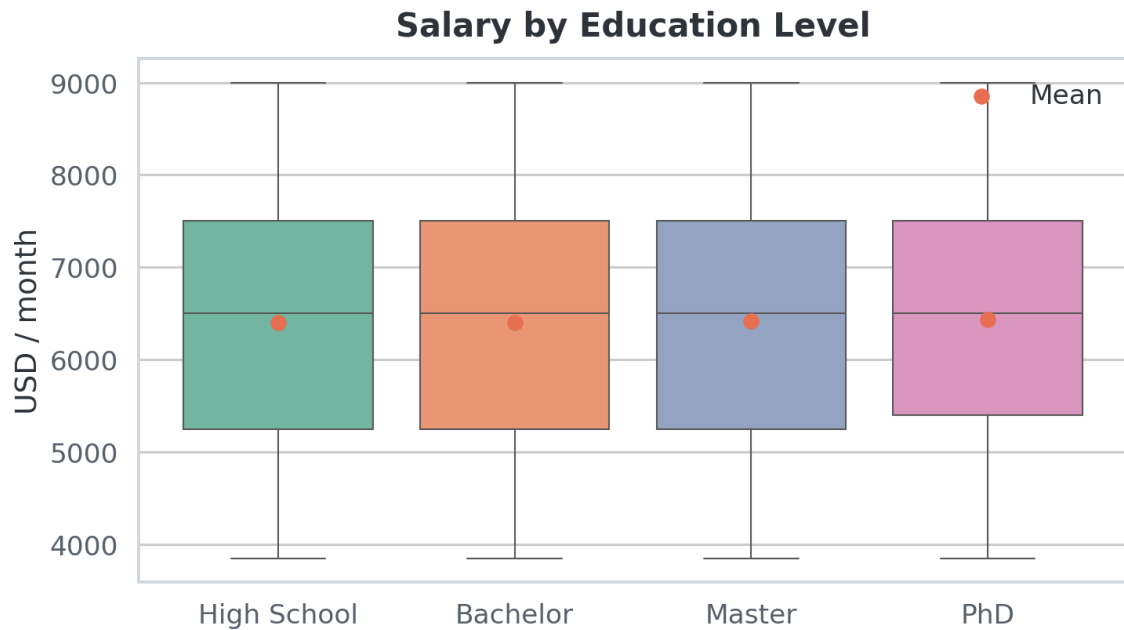
Monthly Salary Distribution



Monthly Salary by Department

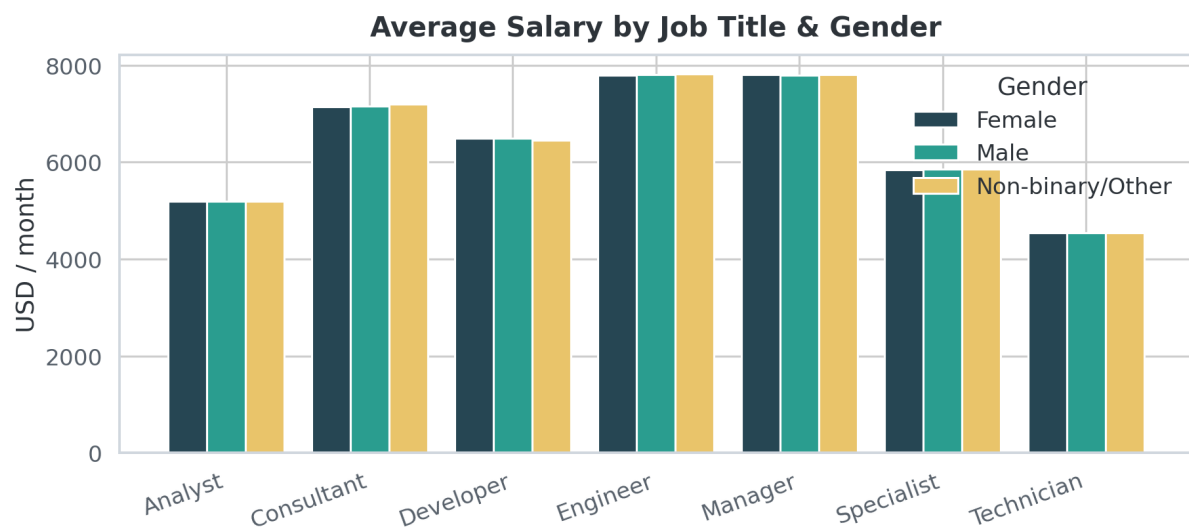


3.1 Salary by education



Education premiums are small: PhD and Master holders earn only modestly above Bachelor graduates, and High-School-educated staff are not far behind. In this workforce, role and measured performance dominate compensation decisions.

3.2 Gender pay equity



Job title	Male – Female (\$/mo)
Analyst	+5.3
Consultant	+23.1
Developer	-0.7
Engineer	+5.5
Manager	-13.9
Specialist	+12.9
Technician	-1.0

Company-wide, women earn \$4.50/month more than men on average; within every job title the largest absolute gap is \$23.10 (Consultant). At an average salary of \$6,403 these differences are economically negligible and consistent with sampling noise. **Conclusion: no gender pay gap to remediate; maintain quarterly monitoring.**

3.3 What actually drives salary — regression model

An OLS regression on all 13 numeric fields plus department, gender, job title and education (one-hot encoded, 80/20 split) achieves $R^2 = 0.9914$ with MAE = \$95 on the held-out 20% — pay is essentially a formula:

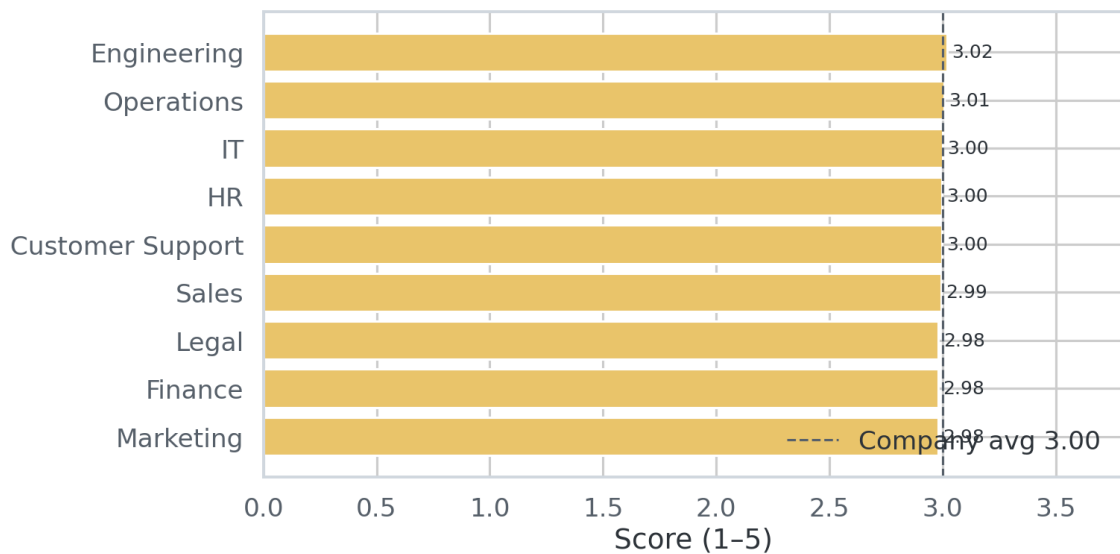
Term	Effect (\$/month)
Job title: Engineer	+2,600
Job title: Manager	+2,599
Job title: Consultant	+1,951
Job title: Developer	+1,300
Job title: Specialist	+650
Job title: Technician	−648
Performance score (per point)	+493
Every other term (dept, gender, education, age, tenure...)	≈ 0 ($ \beta < \3)

Interpretation: compensation governance is transparent and rule-based — a positive property for fairness — but it also means performance measurement quality directly determines \$493/month per point (up to \$1,972 between score 1 and 5). Any bias or noise in performance ratings propagates directly into pay.

4. Performance Analysis

Performance (1–5) averages 3.00 with a near-uniform distribution: 39.9% of staff score 4–5 and 40.1% score 1–2. Departmental averages range only from 2.98 (Finance, Marketing) to 3.02 (Engineering).

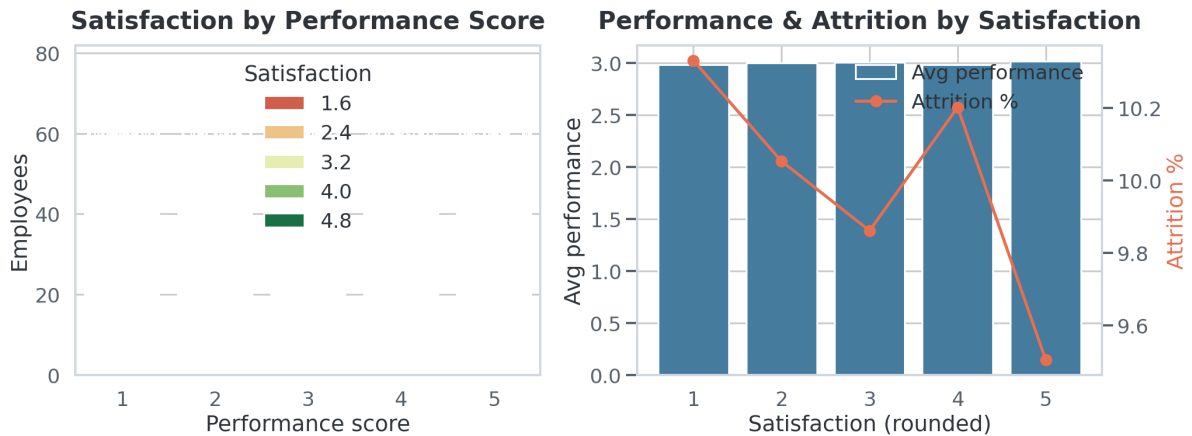
Average Performance Score by Department



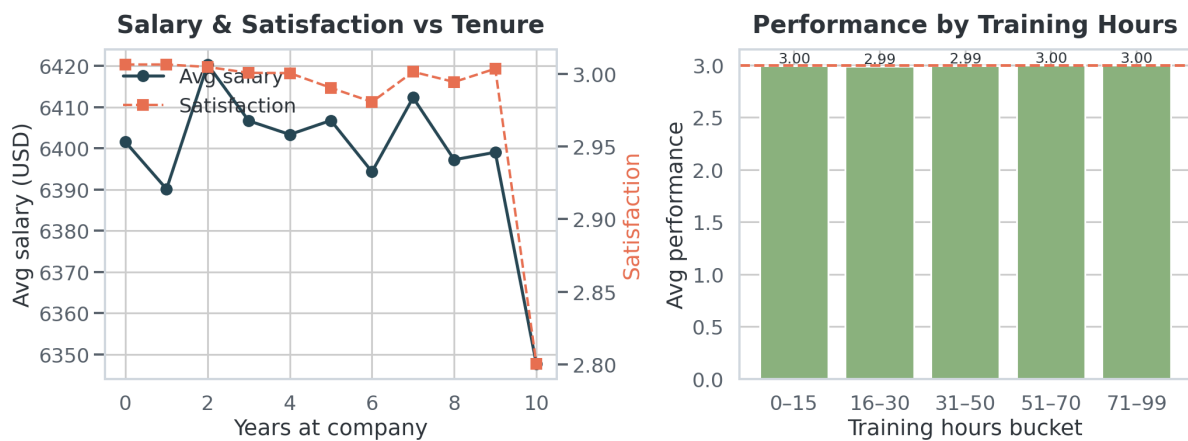
Performance Score Mix by Department (% of staff)



4.1 Performance vs satisfaction and training

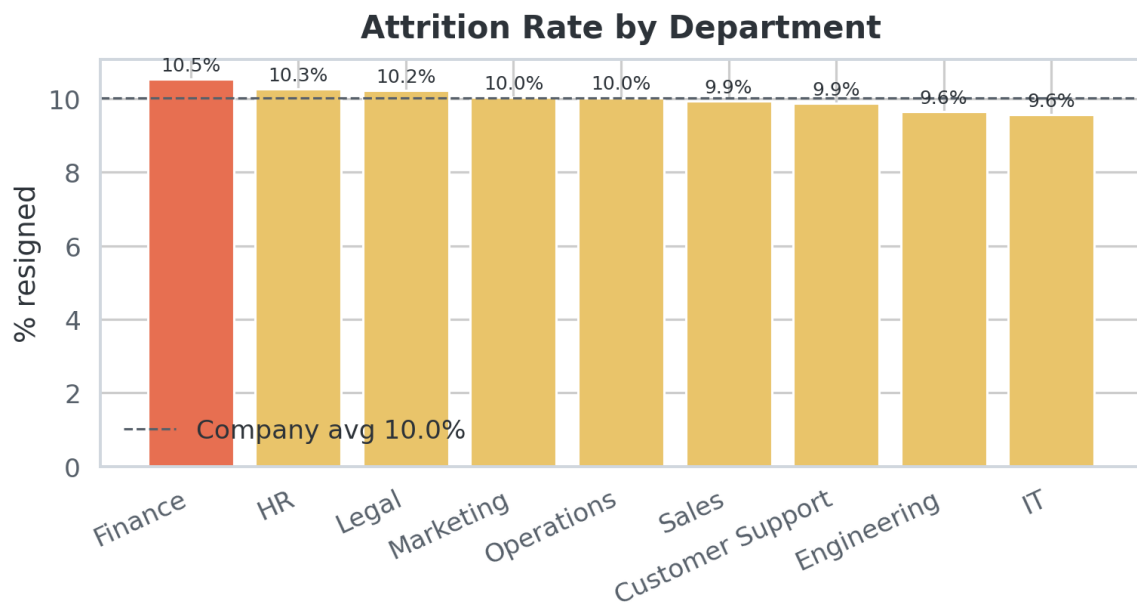


Satisfaction (avg 3.00) is statistically independent of performance ($r = 0.002$) and of attrition. Training hours — 0 to 99, averaging 50 — show no relationship with performance: every training bucket averages 2.99–3.01. In a functioning HR system one would expect at least a mild positive gradient; its total absence is a red flag about how these fields were produced.

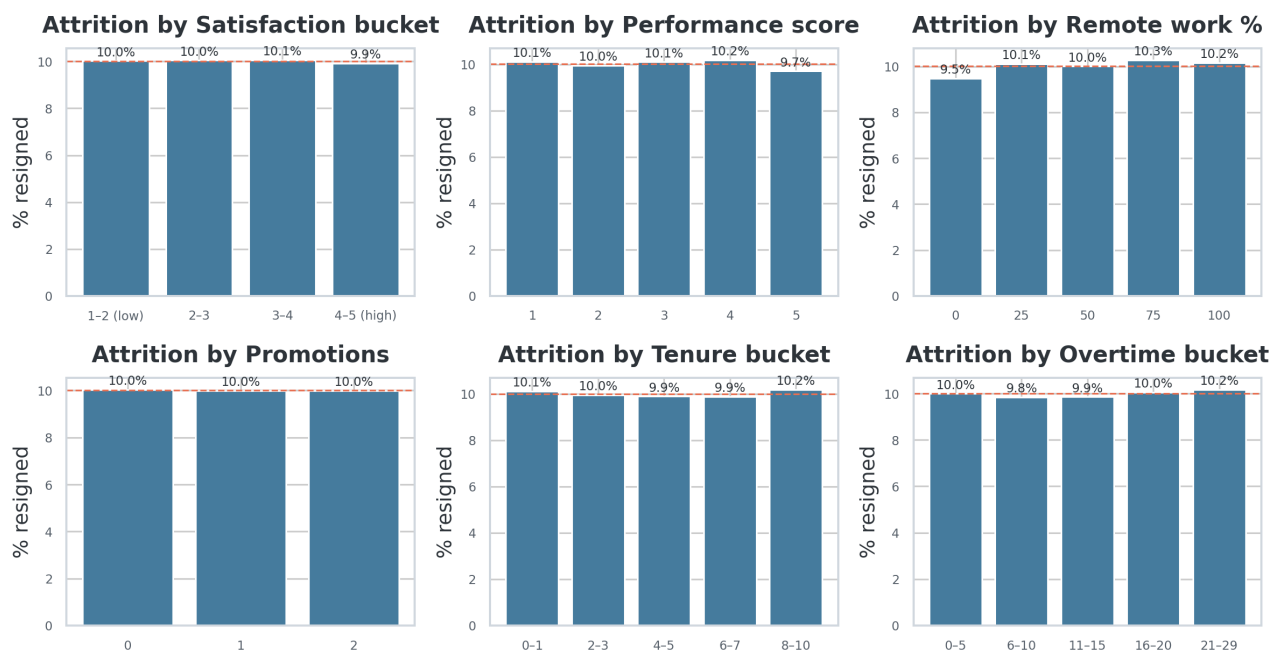


5. Attrition Analysis

Overall attrition is 10.0% (10,010 leavers). The rate is remarkably flat: the widest departmental spread is Finance 10.5% vs Engineering/IT 9.6%.



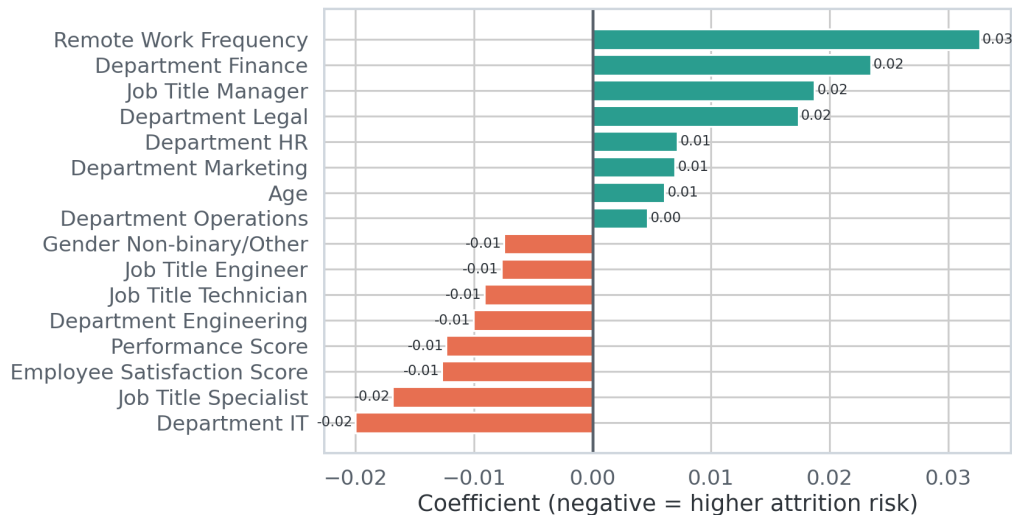
Attrition Drivers — Rate vs. Company Average (dashed line)



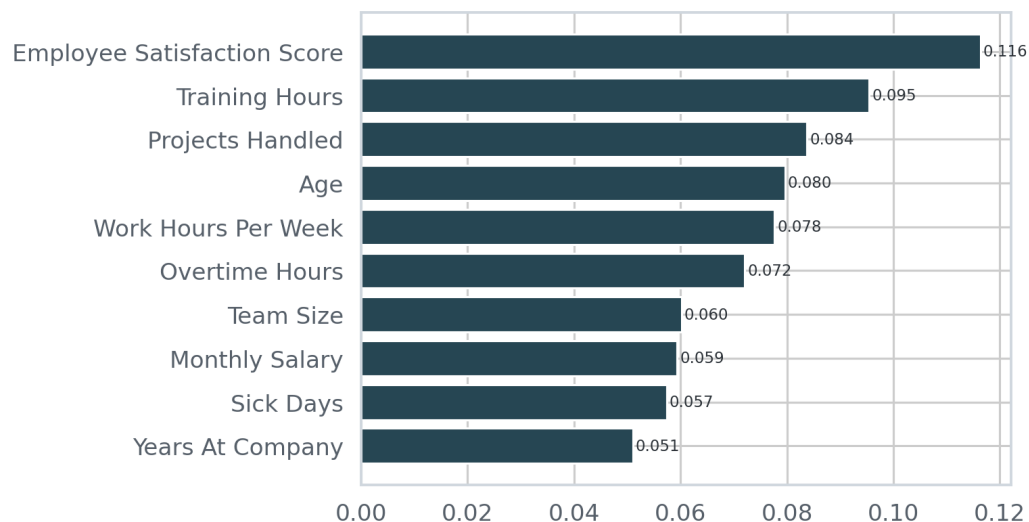
5.1 Predictive modelling — a negative result

Two models were trained on an 80/20 stratified split: logistic regression (accuracy 0.900, AUC 0.490) and a random forest (accuracy 0.900, AUC 0.490). The accuracies look respectable only because the majority class ('stayed') is 90% of the data — always predicting 'stayed' scores identically. The AUC values, which measure genuine discrimination, are ≈ 0.5 : **no better than a coin flip**. Every coefficient is within ± 0.033 and every feature importance below 0.002.

Logistic Regression — Attrition Drivers (standardized coefficients)



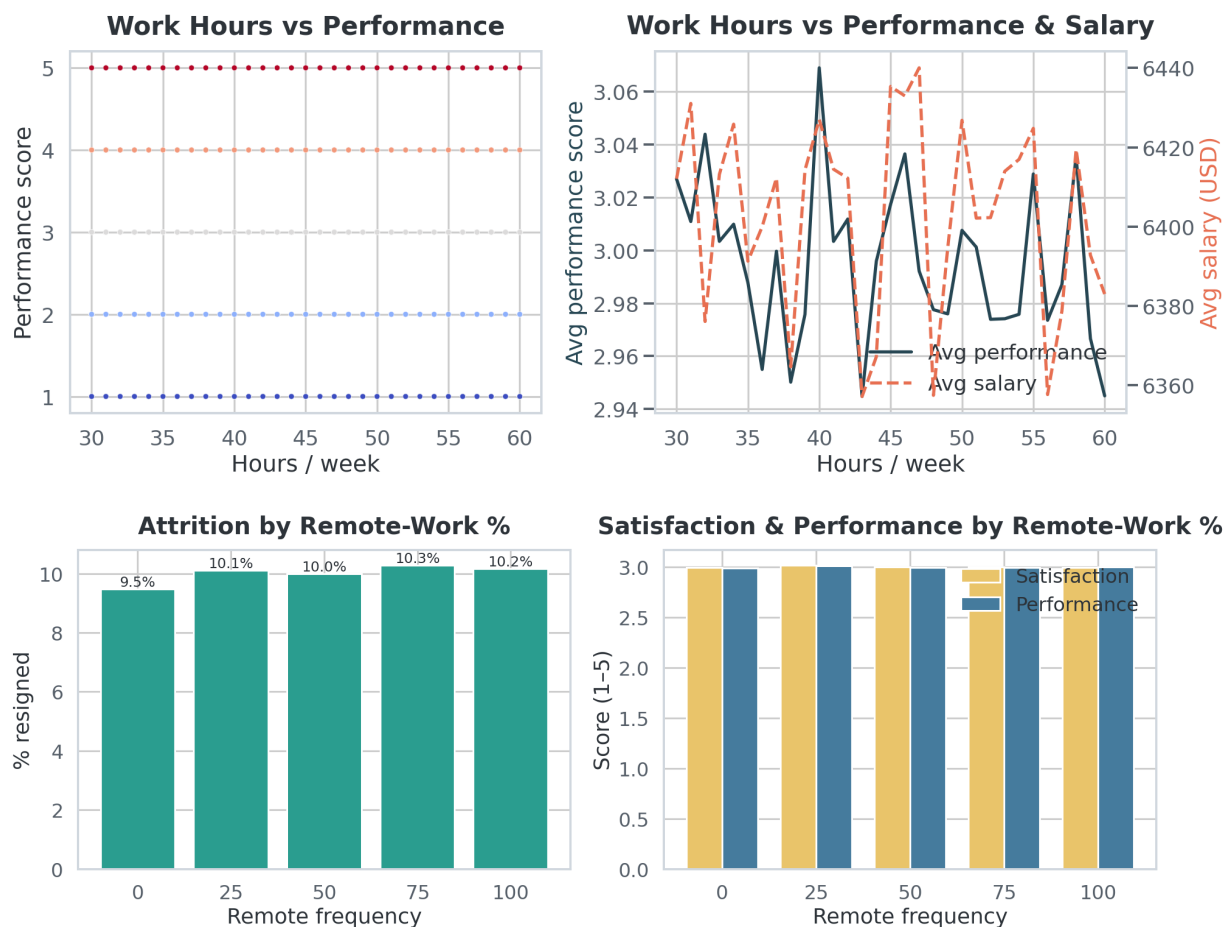
Top Attrition Predictors — Random Forest Importance



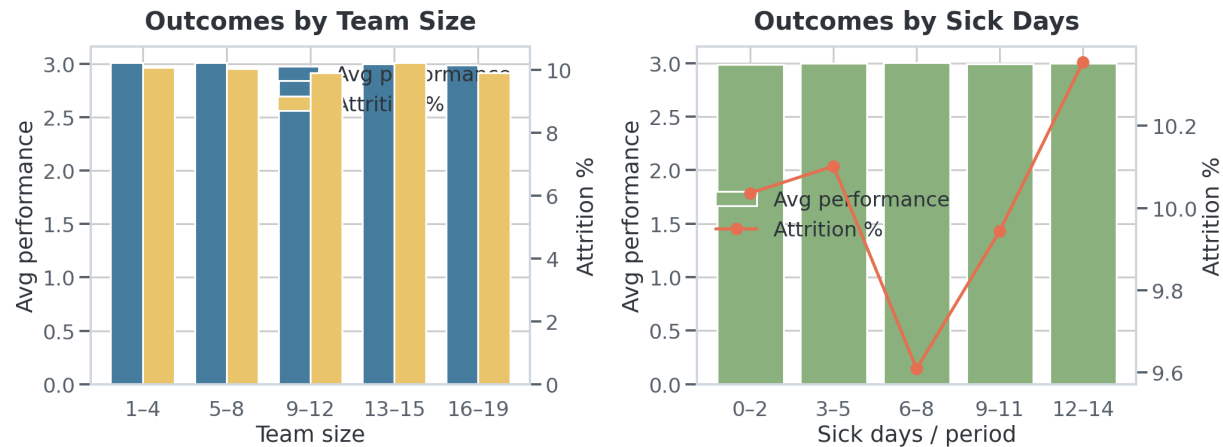
What this means practically: attrition in this dataset is independent of age, tenure, pay, performance, satisfaction, workload, remote policy and promotions. Consequently, no data-driven flight-risk model can be built from the current fields. If this were live organizational data, the correct response would be: (a) audit whether the 'Resigned' flag is populated correctly, (b) enrich the record with exit-interview themes, manager ratings and compensation history, and (c) re-run this pipeline quarterly — it is fully automated.

6. Workload & Work-Life Balance

Staff average 45 hours/week (range 30–60) with 14.5 overtime hours and 7.0 sick days. Neither workload nor remote-work frequency shows any association with performance, satisfaction, or attrition — including the often-debated question of remote work: attrition is 9.5%–10.3% across all five remote-frequency levels.

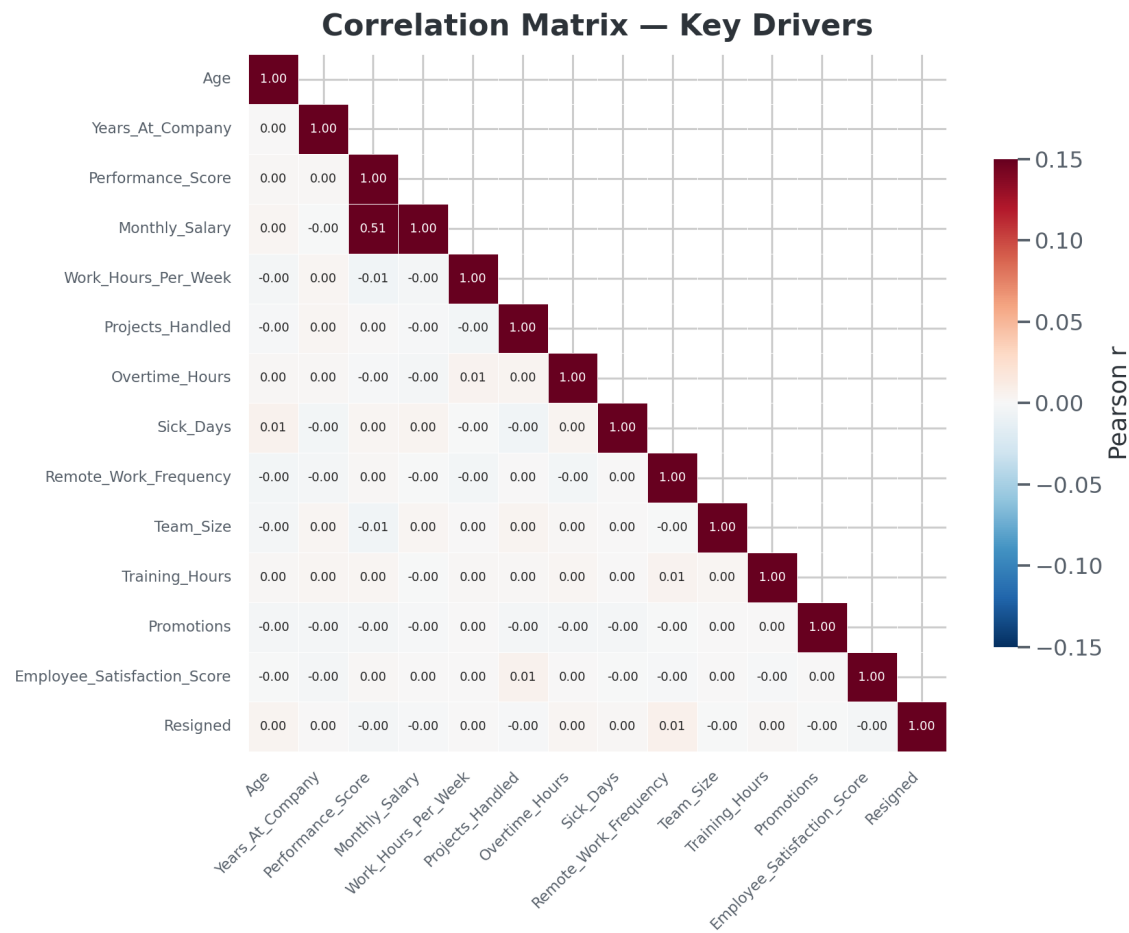


6.1 Team size & sick days



Team size (1–19) and sick days (0–14) are equally inert — performance stays at 3.00 and attrition at ~10% across every bucket. The overtime/sick-day relationship that burnout literature predicts is absent here.

7. Correlation Structure



The correlation matrix contains exactly one meaningful off-diagonal value: salary ↔ performance ($r = 0.51$), explained by the pay formula in section 3.3. All remaining 90 pairwise correlations have $|r| < 0.01$ — a signature of independently generated variables and the strongest evidence that this dataset is synthetic.

8. Recommendations

- 1. Protect the pay-equity position.** Gaps are under 0.4% at every level. Institutionalize the quarterly gap audit (script provided in `src/analysis.py`) so any future drift is caught early.
- 2. Audit performance measurement.** Performance scores drive \$493/month per point but show no correlation with training, workload, projects or tenure — and are uniformly distributed. Verify ratings are evidence-based before expanding their weight in pay.
- 3. Fix attrition data capture.** Attrition is unpredictable from current fields (AUC 0.49). Validate the Resigned flag, add exit-interview categories, manager relationship scores and comp-history, then re-train.
- 4. Don't over-invest in unproven levers.** Training hours, overtime caps and remote-work policies show zero measured effect on outcomes in this data. Run controlled pilots with clear metrics rather than blanket programs.
- 5. Use the pipeline as a standing monitor.** The full analysis (EDA → charts → models → dashboard → report → deck) rebuilds in under a minute with two commands; schedule it against production extracts to keep leadership reporting current.

Appendix A — Methodology

Pipeline: pandas 2.2 for loading/EDA, matplotlib/seaborn for 20 charts, scikit-learn 1.6 for models. Attrition models: LogisticRegression (C=0.5, standardized features, max_iter=2000) and RandomForestClassifier (250 trees, depth 8) on 17 features (13 numeric + 4 one-hot categoricals), 80/20 stratified split, random_state=42; metrics: accuracy, ROC AUC. Salary model: OLS on the same feature matrix minus salary itself; metrics: R², MAE. Correlations: Pearson. All code deterministic.

Appendix B — Data dictionary

Field	Type	Range / values
Employee_ID	int	1 – 100,000 (unique)
Department	cat	9 values (IT, Finance, HR, ...)
Gender	cat	Male, Female, Other
Age	int	22 – 60
Job_Title	cat	7 values (Analyst ... Technician)
Hire_Date	date	2014-09 – 2024-09
Years_At_Company	int	0 – 10
Education_Level	cat	High School, Bachelor, Master, PhD
Performance_Score	int	1 – 5
Monthly_Salary	float	\$3,850 – \$9,000
Work_Hours_Per_Week	int	30 – 60
Projects_Handled	int	0 – 49
Overtime_Hours	int	0 – 29
Sick_Days	int	0 – 14
Remote_Work_Frequency	int	0, 25, 50, 75, 100 (%)
Team_Size	int	1 – 19
Training_Hours	int	0 – 99
Promotions	int	0 – 2
Employee_Satisfaction_Score	float	1.0 – 5.0

Resigned	bool	True 10.01% / False 89.99%
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